WORKSHEET

The Bernoulli distribution

1 Which of the following is an example of a Bernoulli trial? Circle the correct answer.

A Rolling a die and noting the score

B Flipping two coins and recording the outcome

C Drawing a single card from a standard 52 card deck

D Rolling a die and noting if the score is a multiple of 3 or not

E Dealing a 5 card hand from a standard deck of cards.

2 Which of the following is *not* an example of a Bernoulli trial? Circle the correct answer.

A Randomly drawing a single card from a standard deck and noting if it is a king or not

B Flipping a coin and noting the outcome

C Randomly dealing five cards from a standard deck and recording if there are five diamonds

D Rolling a die and noting if the score has a factor of two

E Flipping a coin twice and noting the outcome

3 Determine if the following are examples of Bernoulli experiments. Circle the correct answer.

a Noting the number of vehicles passing over a bridge. Yes/No

b Recording if a car is blue or not as it drives past. Yes/No

c Recording the position of a randomly chosen athlete in a 100 m sprint. Yes/No

d Noting if a person has ever been rock climbing. Yes/No

e Noting if your family has more than three children. Yes/No

f Recording the time it takes to travel to school. Yes/No

4 Calculate the probability of success in the following Bernoulli trials.

a Drawing a red card from a standard deck

b Drawing an Ace from a standard deck

c Drawing the three of clubs from a standard deck

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d Flipping an unbiased coin twice and getting heads both times

e Rolling two dice and the sum of their scores is seven

f Rolling two dice and the product of their scores is six

5 Find the probability of failure in the following Bernoulli trials.

a Rolling a four using a ten-sided die

b Drawing a red jack from a standard deck of cards

c Flipping an unbiased coin three times and getting three tails

d Flipping an unbiased coin three times and getting exactly two heads

e Rolling two six-sided dice and the sum of their scores is eight

f Rolling two 4-sided dice and the difference of their scores is two

6 A batch of 500 LED lights is known to have 8 defective units. If one LED light is chosen at random, and success is selecting a defective unit, what is:

a the probability of success

b the probability of failure.

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Answers

1 D

2 E

3 **a** No

b Yes

c No

d Yes

e Yes

f No

4 **a** $\frac{1}{2}$

b $\frac{1}{13}$

c $\frac{1}{52}$

d $\frac{1}{4}$

e $\frac{1}{6}$

f $\frac{1}{9}$

5 **a** $\frac{9}{10}$

b $\frac{25}{26}$

c $\frac{7}{8}$

d $\frac{5}{8}$

e $\frac{31}{36}$

f $\frac{3}{4}$

6 **a** 0.016

b 0.984

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